

Bitwise differential trail through **ARADI** found by SAT

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1 ARADI

Maximal differential probability: 2^{-32}

0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	4	2	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	4	2	0	0	0	0	0	0	0	0
ARADI ← round0																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	0	0	0	0	0	0	0	0	0	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	0	0	0	0	0	0	0	0	0	
ARADI ← round1																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	4	1	0	0	0	0	0	0	0	0	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	4	1	0	0	0	0	0	0	0	0	
ARADI ← round2																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	4	1	2	8	2	6	1	0	0	0	0	0	0	0	0	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	4	1	2	8	2	6	1	0	0	0	0	0	0	0	0	
ARADI ← round3																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	6	6	9	7	c	7	6	b	0	0	0	0	0	0	0	0	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	6	6	9	7	c	7	6	b	0	0	0	0	0	0	0	0	
FinalRK																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	6	6	9	7	c	7	6	b	0	0	0	0	0	0	0	0	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	6	6	9	7	c	7	6	b	0	0	0	0	0	0	0	0	

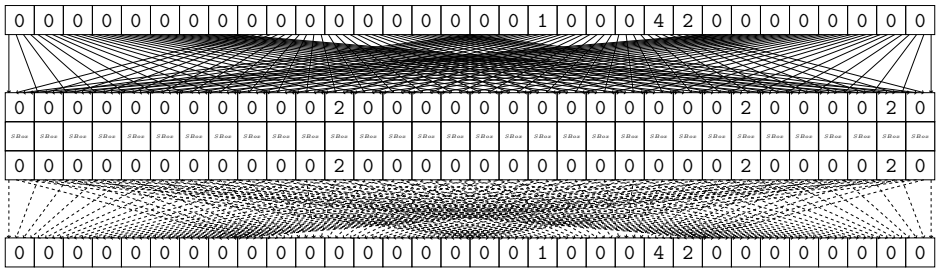
2 ARADI-round0

Maximal differential probability: $2^{-6.0}$

0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	4	2	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	4	2	0	0	0	0	0	0	0
RR																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	4	2	0	0	0	0	0	0	0	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	4	2	0	0	0	0	0	0	0	
SBox Layer																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	4	2	0	0	0	0	0	0	0	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	4	2	0	0	0	0	0	0	0	
Linear Layer 0																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	0	0	0	0	0	0	0	0	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	0	0	0	0	0	0	0	0	

3 SBoxLayer

Maximal differential probability: $2^{-6.0}$



4 LinearLayer0

Maximal differential probability: $2^{-0.0}$

0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	4	2	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	4	2	0	0	0	0	0	0	0	0
xor								xor								xor								xor								
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	0	0	0	0	0	0	0	0	0	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	0	0	0	0	0	0	0	0	0	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	0	0	0	0	0	0	0	0	0	

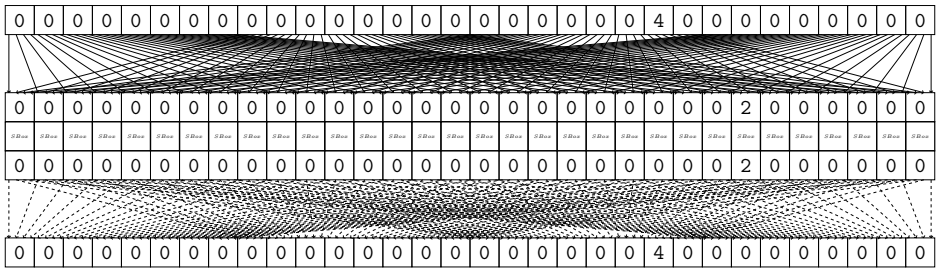
5 ARADI-round1

Maximal differential probability: $2^{-2.0}$

0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	0	0	0	0	0	0	0	0
RK																													
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	0	0	0	0	0	0	0	0
SBox Layer																													
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	0	0	0	0	0	0	0	0
Linear Layer1																													
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	4	1	0	0	0	0	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	4	1	0	0	0	0	

6 SBoxLayer

Maximal differential probability: $2^{-2.0}$



7 LinearLayer1

Maximal differential probability: $2^{-0.0}$

0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	0	0	0	0	0	0	0	0	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	0	0	0	0	0	0	0	0		
0.5								0.5								0.5								0.5							
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	4	1	0	0	0	0	0	0	0	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	4	1	0	0	0	0	0	0	0	

8 ARADI-round2

Maximal differential probability: $2^{-6.0}$

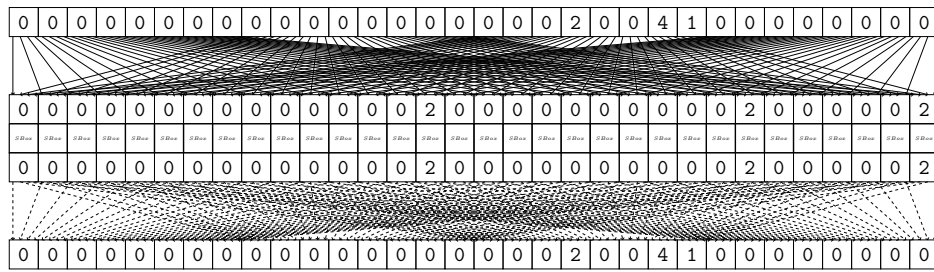
Diagram illustrating a 4-layer neural network architecture. The layers are labeled as follows:

- Input Layer:** 10 nodes (0-9).
- Hidden Layer 1:** 10 nodes (0-9).
- Hidden Layer 2:** 10 nodes (0-9).
- Linear Layer 2:** 10 nodes (0-9).

The diagram shows a grid of nodes with connections between them. The connections are represented by a grid of dots. The first layer has 10 nodes, the second has 10 nodes, the third has 10 nodes, and the fourth has 10 nodes. The connections are shown as a grid of dots between the nodes of adjacent layers.

9 SBoxLayer

Maximal differential probability: $2^{-6.0}$



10 LinearLayer2

Maximal differential probability: $2^{-0.0}$

0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	4	1	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	4	1	0	0	0	0	0	0	0	0
xx								xx								xx								xx							
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	4	1	2	8	2	6	1	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	4	1	2	8	2	6	1	0	0	0	0	0	0	0	0

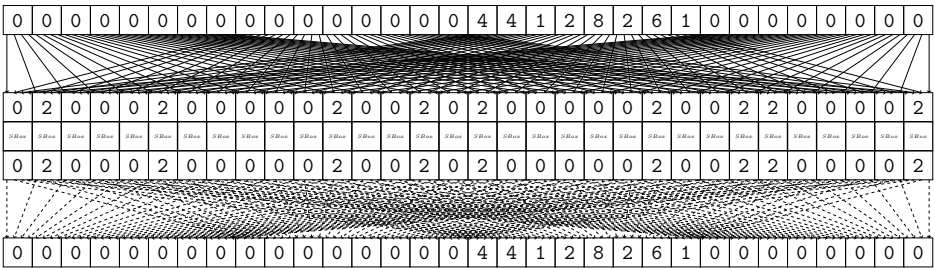
11 ARADI-round3

Maximal differential probability: $2^{-18.0}$

0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	4	1	2	8	2	6	1	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	4	1	2	8	2	6	1	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	4	1	2	8	2	6	1	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	4	1	2	8	2	6	1	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	4	1	2	8	2	6	1	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	4	1	2	8	2	6	1	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	6	6	9	7	c	7	6	b	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	6	6	9	7	c	7	6	b	0	0	0	0	0	0	0

12 SBoxLayer

Maximal differential probability: $2^{-18.0}$



13 LinearLayer3

Maximal differential probability: $2^{-0.0}$

0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	4	1	2	8	2	6	1	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	4	1	2	8	2	6	1	0	0	0	0	0	0	0	0
00								00								00								00							
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	6	6	9	7	c	7	6	b	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	6	6	9	7	c	7	6	b	0	0	0	0	0	0	0	0